CORRIDEX LOGISTICS

Operations & Intelligence Division — Internal Brief

CONFIDENTIAL

OPERATION LAST MILE

Delivery Performance Failure & Supply Chain Risk Investigation

Route Analysis | Warehouse & Stock Investigation | Driver Performance | Customer Impact

DIVISION	DATA PERIOD	SHIPMENTS	ON-TIME RATE	SLA THRESHOLD
Ops & Intelligence	Jan 2024 — Dec 2025	~15,000 Shipments	71.3%	85% | Client: Hartwell Group

— BACKGROUND

Corridex Logistics is a pan-European freight and parcel delivery company operating across 12 cities in 6 countries. Founded in 2018, Corridex has grown from a regional courier into a mid-market logistics provider handling over 15,000 shipments annually across Standard, Express, and Critical priority tiers. The company operates 5 fulfilment warehouses — in Amsterdam, Berlin, Madrid, Warsaw, and Lyon — and serves clients across Retail, Manufacturing, Pharmaceutical, and E-Commerce sectors.

Corridex competes directly with established carriers on two dimensions: price and reliability. Its growth strategy depends entirely on consistently meeting contractual SLA commitments — the promise that shipments arrive within the agreed delivery window.

— THE INCIDENT

Hartwell Group is Corridex's largest client, accounting for 23% of total annual shipment volume. In January 2025, Hartwell Group issued a formal SLA breach notice following a review of their 2024 shipping data. Their logistics audit recorded a Corridex on-time delivery rate of 68.4% against a contractual SLA threshold of 85% — a gap of 16.6 percentage points.

Hartwell Group has invoked Clause 14.3 of their Master Services Agreement. If Corridex does not demonstrate a sustained on-time rate above 85% within two consecutive quarters, Hartwell Group will terminate the contract and transition their volume to a competitor. The contract is valued at EUR 4.2 million annually.

> *“We are not in the business of excuses. Hartwell Group is watching every shipment. I need to know exactly where this operation is failing, which levers to pull, and in what order. Find the signal in the data.”*

— Chief Operations Officer, Corridex Logistics

— COMPETING INTERNAL THEORIES — ALL UNVERIFIED

Four senior stakeholders have each proposed a root cause. All four believe they are right. None of them have looked at the data.

OPERATIONS MANAGER
It is specific drivers and high-risk routes. The performance data will show a small group responsible for the majority of delays.

WAREHOUSE HEAD
Stock is not ready when shipments are scheduled. Warehouse delays are adding hours before a vehicle even leaves the facility.

REGIONAL DIRECTOR
This is not a system-wide problem. One or two city pairs are dragging down the overall number. Fix those routes and the SLA recovers.

FINANCE DIRECTOR
Standard-tier shipments are being deprioritised to protect Express and Critical margins. The SLA breach is a pricing and priority problem, not operational.

— YOUR ROLE

You have been brought in as the lead Data Analyst on the Corridex delivery performance investigation. You have been granted access to two years of shipment records exported from five operational systems — each with its own structure, its own quirks, and its own gaps. Your mandate is to load the data into SQL Server, conduct a structured analytical investigation across six deliverables, and present your findings to the executive team.

You are not here to confirm what the stakeholders already believe. You are here to find what the data actually says — even if that contradicts every theory in the room.

— DATA MODEL — FACTS AND DIMENSIONS

You will receive six tables loaded into SQL Server. The fact_shipments table is the grain — one row per shipment. All five dimension tables connect to it.

TABLE	TYPE	GRAIN	SOURCE SYSTEM
fact_shipments	Fact	One row per shipment	ShipmentCore
dim_customer	Dimension	One row per customer	ClientRegistry
dim_driver	Dimension	One row per driver	FleetManager
dim_route	Dimension	One row per route	RouteMapper
dim_warehouse	Dimension	One row per warehouse	WarehouseOps
dim_date	Dimension	One row per calendar date	Generated

— COLUMN DICTIONARY — KEY FIELDS

COLUMN	TABLE	EXAMPLE	NOTES
shipment_id	Fact	SHP-2024-00001	Unique shipment identifier
customer_id	Fact/Dim	CUS-001	Links Fact to dim_customer
driver_id	Fact/Dim	DRV-001	Links Fact to dim_driver
route_id	Fact/Dim	RTE-001	Links Fact to dim_route
warehouse_id	Fact/Dim	WHS-001	Links Fact to dim_warehouse
shipment_date	Fact	2024-03-14	FK to dim_date
promised_delivery_date	Fact	2024-03-17	SLA commitment date

actual_delivery_date	Fact	2024-03-19	NULL if not yet delivered
expected_transit_days	Fact	3	Based on route — verify against actual
delivery_status	Fact	Late	On-Time, Late, Failed
delay_days	Fact	2	NULL if on-time or failed
shipment_category	Fact	Parcel	Parcel, Pallet, Freight
shipment_weight_kg	Fact	14.5	Gross weight of shipment
shipment_value_eur	Fact	320.00	Declared shipment value
priority_tier	Fact	Express	Standard, Express, Critical
stock_ready_flag	Fact	0	1 = ready at dispatch, 0 = not ready
stock_issue_type	Fact	Out of Stock	NULL if no issue / Out of Stock / Damaged / Late Supplier
warehouse_delay_hours	Fact	6	Hours lost in warehouse before dispatch
delivery_attempt_count	Fact	2	Number of delivery attempts made
failed_delivery_reason	Fact	Wrong Address	NULL if delivered / Vehicle Breakdown / Wrong Address / Customer Unavailable
is_returned	Fact	0	1 = returned to warehouse
customer_segment	Dim	Retail	Retail, Manufacturing, Pharmaceutical, E-Commerce
contract_tier	Dim	Gold	Bronze, Silver, Gold, Platinum
sla_threshold_days	Dim	3	Max delivery days per contract tier
driver_experience_years	Dim	4	Years in logistics
vehicle_type	Dim	HGV	Van, Truck, HGV
performance_rating	Dim	3.8	1.0 to 5.0 — historical rating
route_type	Dim	Cross-Border	Domestic, Cross-Border
risk_level	Dim	High	Low, Medium, High — based on historical performance
standard_transit_days	Dim	3	Expected days under normal route conditions
region	Dim	Western Europe	Northern / Western / Southern / Eastern Europe
is_weekend	Dim	0	1 = Saturday or Sunday
is_public_holiday	Dim	0	1 = public holiday

— DATA QUALITY ISSUES — REQUIRES CLEANING IN SQL

→ actual_delivery_date — NULL for approximately 4% of shipments. These are undelivered or in-transit records. Handle with IS NULL before calculating delay metrics.

→ delivery_status — inconsistent casing across source exports: 'on-time', 'On-Time', and 'ON TIME' all present. Standardise before filtering.

→ warehouse_delay_hours — recorded as 0 for all on-time shipments but NULL for approximately 6% of late shipments. Treat NULL as unknown, not zero.

→ shipment_value_eur — a small subset of records contain negative values from a system reversal error. Exclude from revenue aggregations.

→ delivery_attempt_count — values above 3 exist in the data. These are data entry errors. Cap at 3 or exclude from attempt-based analysis.

→ stock_issue_type — blank where stock_ready_flag = 1. This is expected. Do not treat blank as missing data in these rows.

→ delay_days — pre-calculated column. Verify against actual_delivery_date minus promised_delivery_date before using in analysis.

— ANALYTICAL DELIVERABLES

All six deliverables must be answered in SQL Server after the data has been loaded. Each deliverable maps to a stage of the executive investigation.

#	DELIVERABLE	ANALYTICAL FOCUS	LEVEL
1	**Situation Assessment** Establish the baseline. What does overall delivery performance look like across 2024–2025? Calculate on-time rate by year, quarter, and priority tier. Quantify the gap against the 85% SLA threshold.	Ops & Performance	FOUNDATION
2	**Geographic Breakdown** Is this everywhere or somewhere specific? Break down on-time rate and average delay by origin city, destination city, and route. Identify the worst-performing city pairs and test the Regional Director's theory.	Route Analysis	INTERMEDIATE
3	**Warehouse & Stock Analysis** Are shipments leaving on time? Calculate average warehouse delay hours by warehouse, stock issue frequency, and the relationship between stock_ready_flag and delivery outcome. Test the Warehouse Head's theory.	Supply Chain	INTERMEDIATE
4	**Driver & Route Performance** Who and what is underperforming? Rank drivers by on-time rate, average delay, and failed delivery frequency. Cross-reference with vehicle type and experience level. Test the Operations Manager's theory.	Driver Analysis	INTERMEDIATE
5	**Customer Impact Analysis** Which customers are absorbing the most damage? Calculate on-time rate, total delay days, and failed shipments by customer, segment, and contract tier. Identify whether Hartwell Group is the worst-affected account.	Client Risk	ADVANCED
6	**Delivery Risk Score** Combine route risk level, warehouse delay frequency, driver performance rating, priority tier, and stock readiness into a composite risk score per shipment. Rank the highest-risk shipment profiles.	Risk Intelligence	ADVANCED

— SUBMISSION REQUIREMENTS

#	DELIVERABLE	REQUIREMENT
01	**SQL Script**	One .sql file containing all queries for the six analytical deliverables, clearly labelled with section headers and business context comments.
02	**Views**	At least two named Views created in SQL Server that simplify repeated logic used across multiple deliverables.

03	**Written Commentary**	A short document (max 1 page) stating your key finding, which stakeholder theory the data supports or refutes, and your top recommendation to the COO.

The data reflects real operational patterns. Every cleaning decision must be intentional and defensible.

The answer is in the data.